GATE Data Science and Artificial Intelligence

Topic: Probability and Statistics

Subtopic: Multiplicative law of probability

LECTURE - 11

GATE SMASHERS

EXPLAINATION

Multiplicative Law - If A and B are two events associated with same random experiment. The probability of simultaneous occurence of two events A and B is equal to the probability of one of the two events multiplied by the conditional probability of the other.

$$P(A \cap B) = P(AB) = P(A)\,P(B/A)$$

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Example 1 - Three balls are drawn successively from a box containing 6 red, 4 white and 5 blue balls. Find the probability that these are drawn in order, red white and blue, if each ball drawn is not replaced.

Example 2- From a pack of 52 cards, 4 cards are drawn one by one without replacement. Find the probability that all are aces.

Example 3 - What is the probability of drawing in succession an ace, king and queen of club from the deck of cards under the assumption that the cards are not replaced after each draw.